**10** (a) Simplify $\left(\frac{1}{m^2}\right)^0$

Any nonzero base to power 0 equals 1.

**1 mark**
1

**(1)**

(b) Simplify $\frac{8(x-4)}{(x-4)^2}$

$8(x-4)/(x-4)^2 = 8/(x-4)$.

**1 mark**
$8/(x-4)$

**(1)**

(c) Simplify $(3n^4w^2)^3$

$(3n^4w^2)^3 = 3^3n^{12}w^6 = 27n^{12}w^6$.

**2 marks total**
$27, n^{12}, w^6$

**(2)**

**(Total for Question 10 is 4 marks)**

**11** Jack is in a restaurant.
There are 5 starters, 8 main courses and some desserts on the menu.

Jack is going to choose one starter, one main course and one dessert.
He says there are 240 ways that he can choose his starter, his main course and his dessert.

Could Jack be correct?
You must show how you get your answer.

**(Total for Question 11 is 2 marks)**

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